Discrete Structures

Quiz 1

Q#1 $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r) \equiv$
$(p \wedge q) \vee (\sim p \wedge r)$

Using LHS:

$= (p \wedge q) \vee (\sim p \wedge r) \vee [(q \wedge r) \wedge t]$ . . .

$= (p \wedge q) \vee (\sim p \wedge r) \vee [(q \wedge r) \wedge (p \vee \sim p)]$

Apply Distributive law

$= (p \wedge q) \vee (\sim p \wedge r) \vee (p \wedge q \wedge r) \vee (\sim p \wedge q \wedge r)$

$= [(p \wedge q) \vee (p \wedge q \wedge r)] \vee [(\sim p \wedge r) \vee (\sim p \wedge r \wedge q)]$

Apply Absorbtion law

$= [p \wedge q] \vee [\sim p \wedge r]$

LHS = RHS

Hence proved.

Q #2 simplify

$$\sim[\sim(p \vee \sim q) \vee \sim(p \vee q)]$$

$A = p \vee \sim q$, $B = p \vee q$

$= \sim[\sim A \vee \sim B]$

apply demorgan's law:

$= A \wedge B$

$= (p \vee \sim q) \wedge (p \vee q)$

$= (p \wedge p) \vee (p \wedge q) \vee (p \wedge \sim q) \vee (q \wedge \sim q)$

$= p \vee (p \wedge q) \vee (p \wedge \sim q) \vee c$

$= p \vee c$

$= p$

Q#3 show

$$(p \wedge \sim q) \vee (p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \equiv p$$

LHS:

Apply distributive law:

$$= (p \wedge \sim q) \vee [(p \wedge q) \wedge (r \vee \sim r)]$$

$$= (p \wedge \sim q) \vee [(p \wedge q) \wedge (t)]$$

$$= (p \wedge \sim q) \vee (p \wedge q)$$

Apply distributive law:

$$= p \wedge (\sim q \vee q)$$

$$= p \wedge (t)$$

$$= p.$$

LHS = RHS,

Hence proved.

Q#4 Simplify

$$(p \wedge q) \vee [p \wedge (q \vee r)] \vee [q \wedge (q \vee r)]$$

Apply absorbtion:

$$= (p \wedge q) \vee (p \wedge (q \vee r)) \vee (q)$$

$$= (p \wedge q) \vee (p \wedge q) \vee (p \wedge r) \vee q$$

$$= (p \wedge q) \vee (p \wedge r) \vee q$$

$$= (p \wedge q) \vee q \vee (p \wedge r)$$

apply absorbtion:

$$\boxed{= q \vee (p \wedge r)}$$

Q #5

" It is not the case that you are either a junior staffer & without a keycard, or that you are not a junior staffer but also do not have key card".

1) Translate into symbols (J for junior staff, K for keycard):

$$= \sim[(J \wedge \sim K) \vee (\sim J \wedge \sim K)]$$

2) Simplify & write in simple english:

$$= \sim[(J \vee \sim J) \wedge \sim K]$$

$$= \sim[t \wedge \sim K]$$

$$= \sim[\sim K]$$

$$= K$$

" You must have a key card".